A MATHEMATICAL MULTISCALE FRAMEWORK COUPLING WITHIN-HOST RESISTANCE EVOLUTION AND BETWEEN-HOST TRANSMISSION

Felipe Schardong[1], Claudio Jose Struchiner[1], Luiz Max Carvalho[1]

School of Applied Mathematics, Getulio Vargas Foundation (FGV - EMAp)

felipe.schardong@fgv.edu.br[1], claudio.struchiner@fgv.br[1], luiz.fagundes@fgv.br[1]

Introduction

Antimicrobial resistance (AMR) is a major global health threat, responsible for an estimated 1.27 million direct deaths in 2019 and projected to cause over 39 million cumulative direct deaths between 2025 and 2050 [1, 2]. Resistance emerges within hosts through drug exposure, pathogen growth, mutation, and horizontal gene transfer [3]. Its spread at the population level depends on transmission and spatial connectivity [4, 5]. Mathematical models have addressed within-host pharmacology [6, 7] and between-host transmission [8, 9], while stochastic approaches have highlighted the role of demographic fluctuations in the establishment of rare resistant strains [10, 11]. However, these processes are often studied separately, limiting our understanding of how drug exposure and stochastic resistance evolution within hosts interact with transmission and spatial connectivity to shape the emergence and spread of AMR. Here, we develop a mathematical multiscale framework coupling pharmacokinetics and pharmacodynamics, stochastic within-host resistance evolution, and lineage-specific between-host transmission in a two-patch metapopulation. The framework combines PBFTPK models [12, 13], pharmacodynamic selection mechanisms [14], and stochastic population dynamics [15, 16]. We use it to investigate how different patterns of treatment non-adherence [17, 18] and asymmetric mobility shape resistance emergence and spread.

Methods

An overview of the general mathematical framework is illustrated in Figure 1, linking drug dosing to population-level spatial spread.

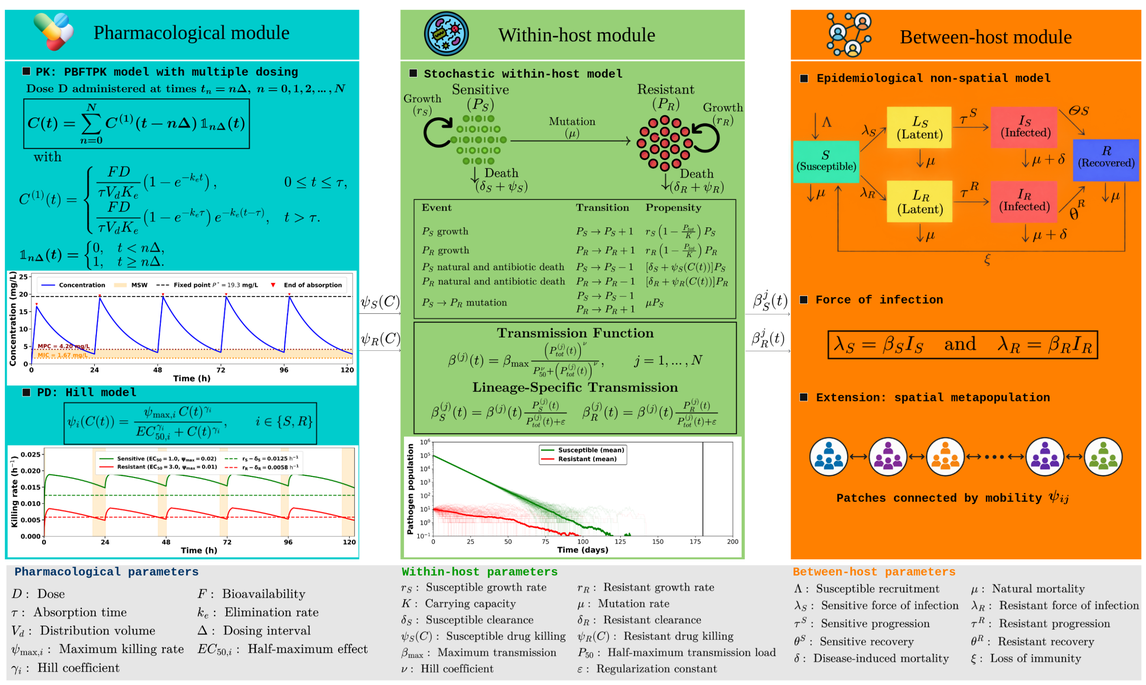


Figure 1: Schematic representation of the mathematic multiscale mathematical framework coupling within-host resistance evolution and between-host transmission. Source: Author.

For each scenario, we define non-adherence sub-levels, summarized in Table 1, to assess how deviations from the prescribed regimen affect resistance development.

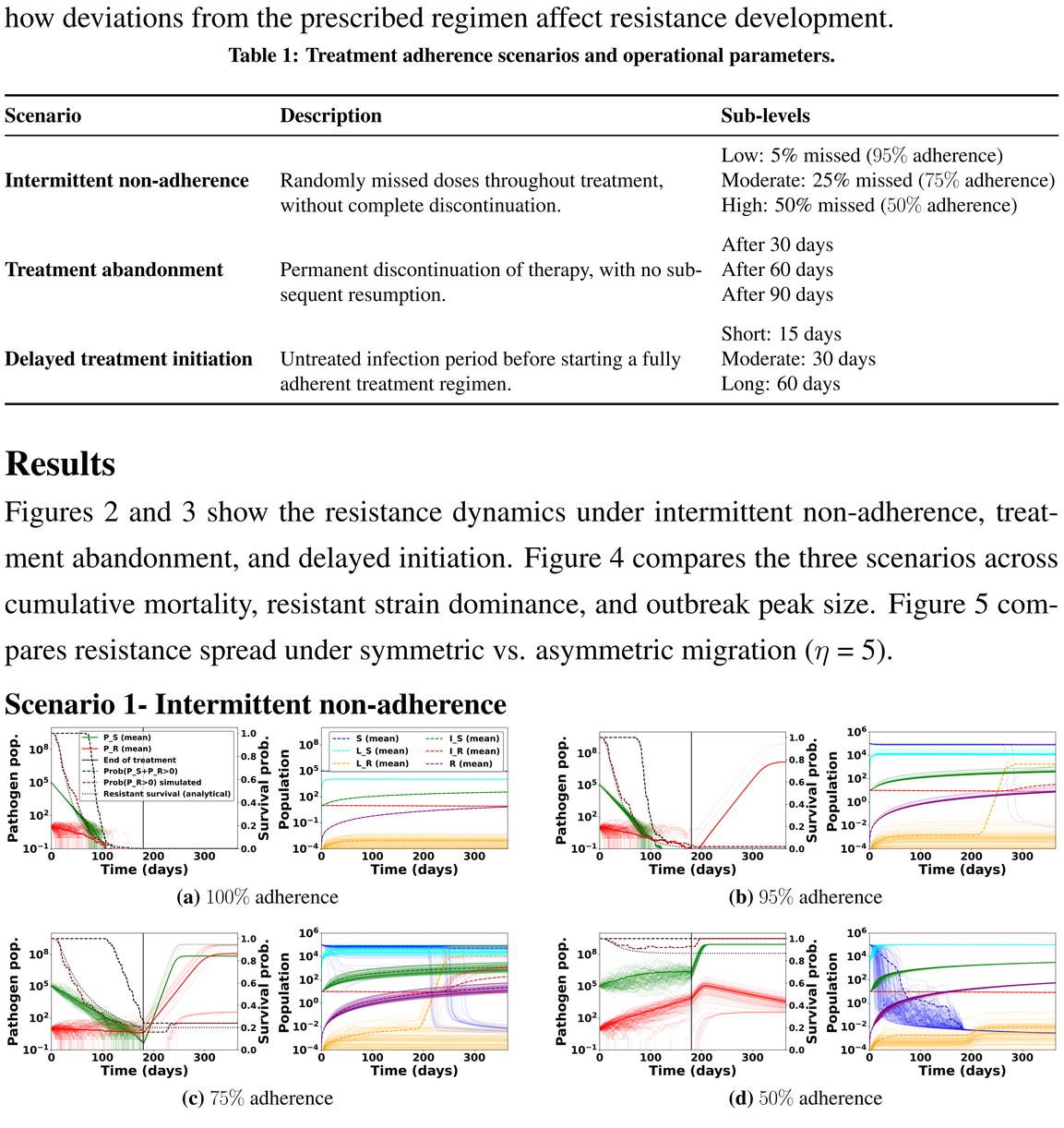
Table 1: Treatment adherence scenarios and operational parameters.

Scenario	Description	Sub-levels
Intermittent non-adherence	Randomly missed doses throughout treatment, without complete discontinuation.	Low: 5% missed (95% adherence) Moderate: 25% missed (75% adherence) High: 50% missed (50% adherence)
Treatment abandonment	Permanent discontinuation of therapy, with no subsequent resumption.	After 30 days After 60 days After 90 days
Delayed treatment initiation	Untreated infection period before starting a fully adherent treatment regimen.	Short: 15 days Moderate: 30 days Long: 60 days

Results

Figures 2 and 3 show the resistance dynamics under intermittent non-adherence, treatment abandonment, and delayed initiation. Figure 4 compares the three scenarios across cumulative mortality, resistant strain dominance, and outbreak peak size. Figure 5 compares resistance spread under symmetric vs. asymmetric migration ($\eta = 5$).

Scenario 1- Intermittent non-adherence

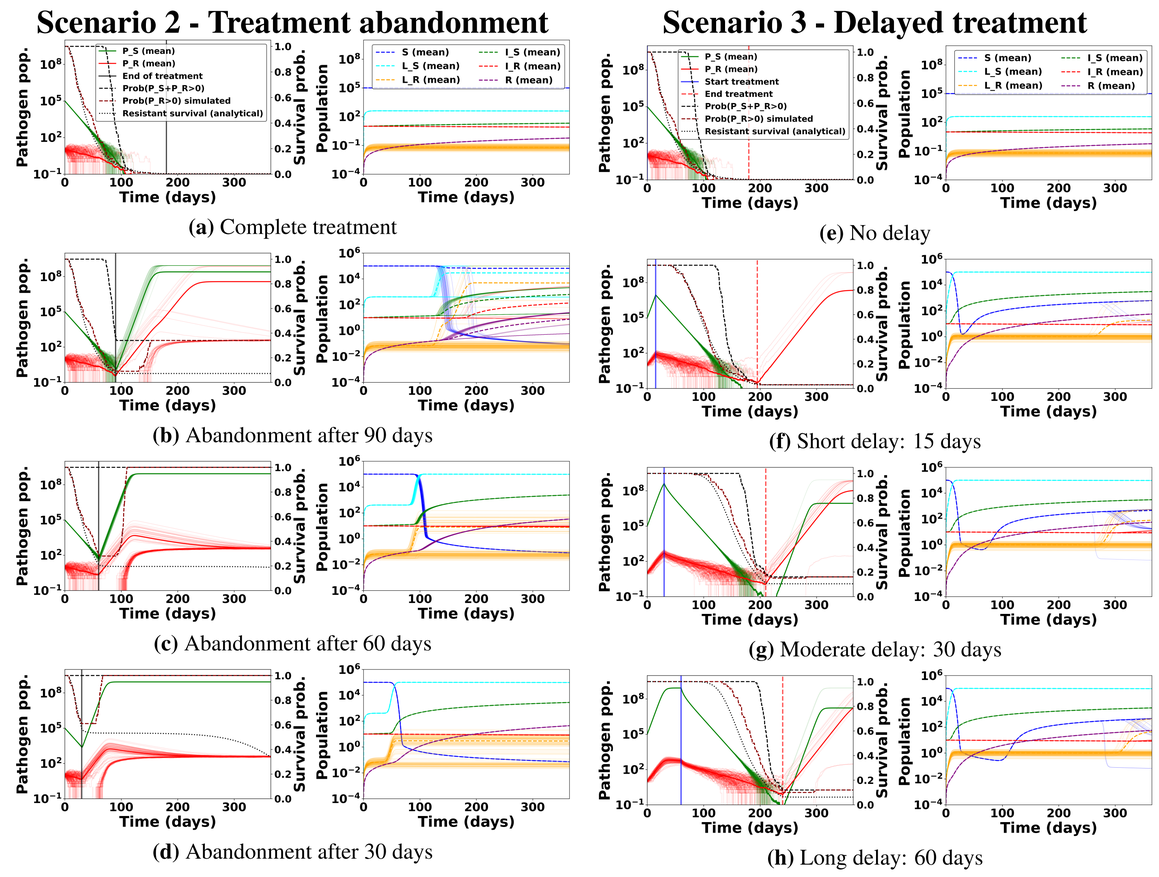


(a) 100% adherence **(b)** 95% adherence **(c)** 75% adherence **(d)** 50% adherence

Figure 2: Adherence effects at 100%, 95%, 75%, and 50% on multiscale resistance dynamics. Panels $a - d$ correspond to the four adherence rates. Source: Author.

Scenario 2 - Treatment abandonment

Scenario 3 - Delayed treatment



(a) Complete treatment **(e)** No delay

(b) Abandonment after 90 days **(f)** Short delay: 15 days

(c) Abandonment after 60 days **(g)** Moderate delay: 30 days

(d) Abandonment after 30 days **(h)** Long delay: 60 days

Figure 3: Resistance dynamics under treatment abandonment (a–d) and delayed initiation (e–h). Pathogen population (left axis) and host subpopulations (right axis). Source: Author.

Synthesis of treatment non-adherence scenarios

The effects of non-adherence are nonlinear. Small lapses have limited effects, but lower adherence or early treatment interruption can lead to outbreaks, while delayed treatment leads to the largest outbreaks.

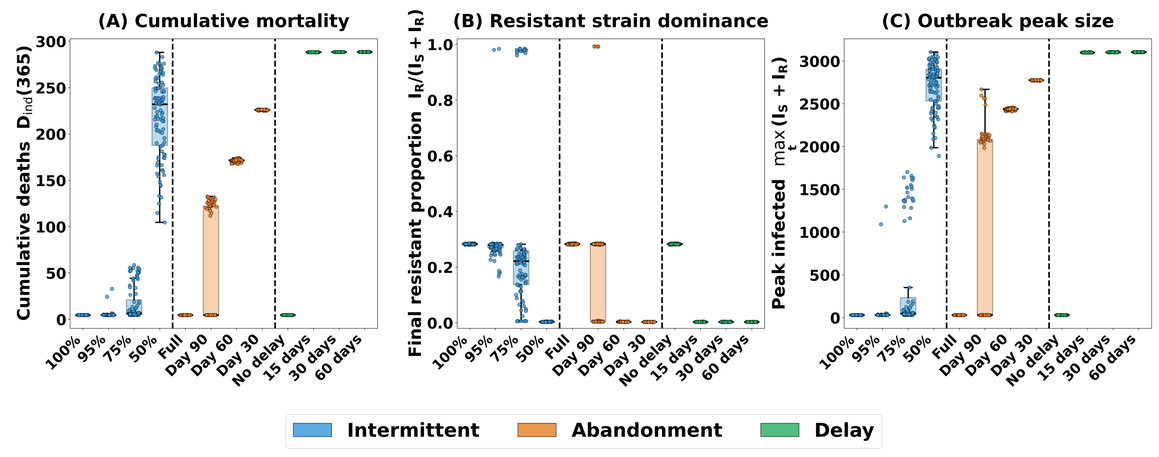
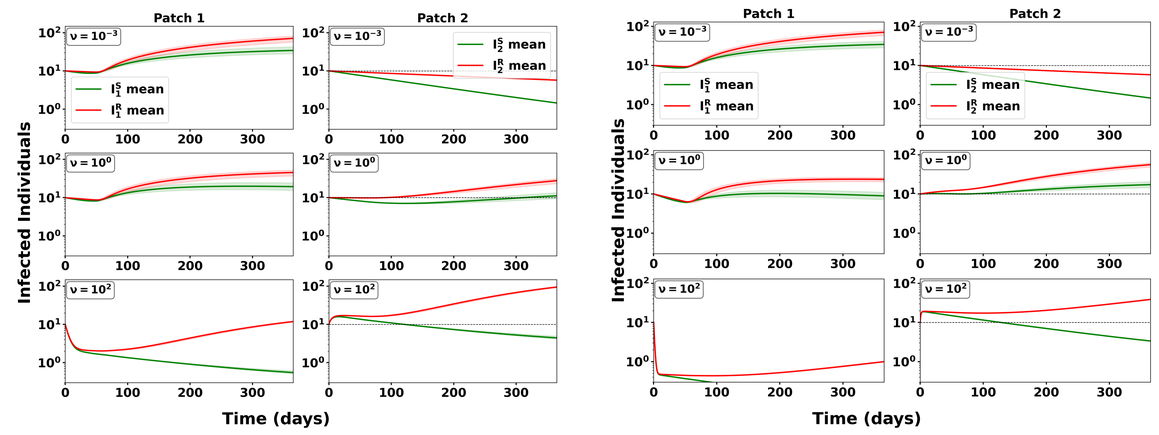


Figure 4: Comparative synthesis of treatment non-adherence scenarios. Source: Author.

Spatial spread: symmetric vs. asymmetric migration

Resistance rescue in Patch 2 is rare at low connectivity ($\nu \le 10^{-1}$) but nearly certain from baseline connectivity onward ($\nu \ge 10^{0}$). Asymmetric infectious outflow ($\eta = 5$) lowers this migration threshold and amplifies the resistant burden at intermediate connectivity.



(a) Homogeneous (symmetric) migration **(b)** Asymmetric migration ($\eta = 5$)

Figure 5: Migration symmetry shapes resistance spread under heterogeneous adherence: (a) homogeneous vs. (b) asymmetric migration ($\eta = 5$). Source: Author.

Conclusions

- Treatment non-adherence increases mortality and outbreak size, with severe effects at adherence levels $\le 50\%$ or early treatment interruption (≤ 60 days).
- Delayed treatment initiation (≥ 15 days) leads to the worst outcomes, with further delays having limited additional effects.
- The decline in the resistant fraction (Fig. 4 - B) reflects dilution by susceptible strain expansion rather than a reduction in the absolute resistant load.
- Asymmetric migration promotes resistant strain spread from non-adherent hotspots, enabling stochastic rescue and late outbreaks in adherent communities.
- Effective AMR control requires adherence support and targeted infection control in hotspots and surrounding communities.

References

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